Worksheet 2-10 A Derivatives of Other Trig Functions #1
AP Calculus Name_____

Directions: Find the derivative of each function.

1. $f(\theta) = \dfrac{\sec\theta}{\theta}$

2. $g(x) = \csc x + \ln x$

3. $y = 5x - \cot x$

4. $h(x) = 3x \sec x$

5. $p(x) = \csc x \cos x$

6. Given $f(x) = \sec x$, find the equation of the tangent line to the graph of $f(x)$ at $x = \dfrac{\pi}{4}$.

7. Given $g(x) = 3\csc x$, find the instantaneous rate of change of $g(x)$ at $x = -\frac{\pi}{6}$.

Directions: Estimate the derivative at the given *x*-value using a calculator. Remember to round all final answers to three decimal places.

8. $h(x) = -2\sec x^3$ at $x = 3$

9. $f(x) = \tan^2(3x)$ at $x = -2$